

Generator Remote Start & Automation

PREPARED FOR Dilanka	SITE Polonnaruwa	DATE 13 September 2026	REFERENCE KT-DSE710-001
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1. SUMMARY

Your 35 kVA standby generator needs a person at the panel to start it, and its battery must be disconnected between runs. We propose to make the set start on command from a phone or computer, keep its battery charged and ready, and report its own status — using the automatic start controller *already fitted* to your machine.

2. THE SITUATION TODAY

Starting the generator takes five manual steps: reconnect the battery, reset the controller, select manual mode, press start, then throw the changeover.

Disconnecting the battery is not carelessness — it is currently necessary. The 710 controller draws roughly 150 mA continuously — about 25 Ah a week, which flattens the starting battery in about a fortnight. With the charger clipped on by hand rather than permanently wired, disconnecting is the only thing protecting it.

That same disconnection blocks automation: a controller with no power cannot receive a start command. Restoring permanent, charged power is the foundation of this work, and is in scope.

3. WHAT YOU WILL GET

- **Start and stop the generator remotely** from a phone or web browser
- **No more disconnecting the battery** — permanently connected and charged
- **Automatic weekly test run** — keeps a standby engine healthy and proves it starts
- **Automatic alerts** — running, failed to start, controller alarm, unit offline (section 9)

4. SCOPE OF WORKS

Battery & charging

- **Permanently wire your existing UPE 24-10A charger** on its own breaker, replacing the clip-on leads — subject to the float-voltage check below
- Fused, properly lugged battery cabling and a 250 A isolator, retained for maintenance isolation only
- Test the batteries and the engine's charging alternator

Controls & monitoring

- Configure the existing 710 auto start controller for automatic operation
- Build and install the isolated remote start control unit
- Commission to the manufacturer's published test procedure
- Connect to the Ktronics platform; configure alerts and the weekly test run
- Handover and operating instructions for your staff

Charger check. A permanently connected charger must settle to a float voltage of about 27.2–27.6 V. We will confirm your UPE unit does; if it holds a higher voltage indefinitely it would gas the batteries dry, and we would quote a replacement rather than wire it in.

5. WHAT STAYS MANUAL — AND WHY

The changeover from mains to generator remains manual, by design. Automatic transfer without a properly interlocked switchboard can push generator power back into the CEB network. That is dangerous to line workers and not permitted. Your operator continues to throw the changeover once the set is running and warmed up.

Fully automatic transfer can be added later with an interlocked transfer switch, quoted separately — this design makes that upgrade an addition, not a rebuild.

6. TIMELINE

Site survey and testing (1 day) → battery, charger and cabling works (1 day) → **7-day settling period** → automation, commissioning and handover (1 day). The settling period is deliberate: we confirm the set holds charge unattended before automating it.

7. ASSUMPTIONS AND EXCLUSIONS

Assumed

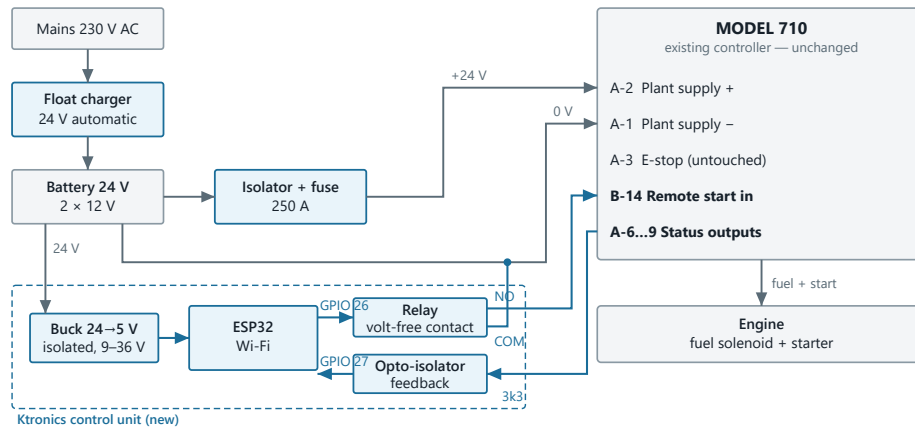
- Mains supply point at the generator for the charger
- Wi-Fi or mobile data coverage at the generator
- Engine mechanically sound and cranks reliably

Not included

- Replacement batteries or charger, if testing shows either unserviceable
- Engine, fuel, exhaust and charging alternator repairs
- Automatic transfer switch and switchboard interlocking

8. TECHNICAL DESIGN

The 710 controller starts the set when its **remote start input (connector B, terminal 14)** is connected to plant supply negative. Our control unit provides exactly that, as an **isolated, volt-free contact** — so no Ktronics equipment becomes part of the generator's control circuit, and its protections and timers are unchanged.



Grey = existing plant. Blue = supplied by Ktronics. The only connection into the controller is a volt-free contact across B-14 and A-1.

9. INTERFACE AND PIN MAPPING

Control unit (ESP32)

PIN	FUNCTION
VIN / 5 V	Supply from isolated buck converter
GND	Unit ground — isolated from battery –
GPIO 26	Output → relay driver (remote start)
GPIO 27 / 14	In ← engine running, common alarm
GPIO 25 / 33	In ← system in auto , fail to start

Generator side (710 controller)

TERMINAL	CONNECTION
B-14	Remote start input → relay contact NO
A-1	Plant supply negative → relay contact COM
A-6 ... A-9	Four configurable outputs → opto-isolators: engine running, common alarm, system in auto, fail to start
A-2, A-3	Supply positive and emergency stop — untouched

Design notes

- Fail-safe at power-up.** GPIO 26 carries a 10 kΩ pull-down, so the relay cannot close while the ESP32 boots or resets — a rebooting unit can never start your set.
- Maintained signal.** The contact stays closed for the whole run; opening it begins the controller's own cooldown and stop sequence. Nothing is pulsed or forced.
- Local control always wins.** If the unit or network fails, the set behaves as it does today and stays startable from the panel. No protection is bypassed — alarms, shutdowns and the emergency stop remain the controller's authority.
- It knows when it is not ready.** The controller reports *system in auto*, so being left out of automatic mode raises an alert instead of failing silently. Hold-up capacitance keeps it powered through cranking voltage sag.

How the alerts work

ALERT	DETECTED BY
Started / stopped	Controller's <i>engine running</i> output changes state
Failed to start	Controller's <i>fail to start</i> output, backed by a 90 s timeout
Controller alarm	Common alarm — oil, temperature, battery, charge failure, e-stop
Not in auto	<i>System in auto</i> output lost — set will not accept a remote start
Unit offline	Platform stops receiving the 5-minute heartbeat

Every alert is a real signal off the controller, not an inference.

10. ACCEPTANCE

Commercial terms and pricing will be issued separately. To proceed on scope, please sign below and return a copy to Ktronics.

For and on behalf of the Customer · Name, position, date

For and on behalf of Ktronics (Pvt) Ltd · Name, position, date